AP PHYSICS 1 — IA-1: LEARN BY QUESTIONS

Exam Tue 27 Oct, 9:40 AM. 30 marks, 2 hours, no calculator.

Method: 60 questions in 8 chains. Every question unlocks the next. Never memorise an answer — only a technique.

HOW TO USE THIS TONIGHT (4 hours):

1. Read the question. Say your answer OUT LOUD in one sentence.

2. Read the technique box. Say it back.

3. Only THEN read the worked answer.

4. Jump to the next chain. Tomorrow morning: re-say only the technique boxes.

CHAIN 1 — MEASUREMENT (45 min)

Q1. You measure a rod with a ruler. Why does your number ever differ from the true value?

> *You answered something like "because the ruler is bad". Better: measurement always has a reading + a unit, and every reading carries an uncertainty from the instrument and the reader.*

TECHNIQUE A — the three error families. Write these three words and their one-line difference. That IS the definition question:

- **Random error** = unpredictable, up-and-down scatter (reader parallax, vibration). Reduces by averaging many readings.
- **Systematic error** = one-way, consistent bias (zero error, worn scale, wrong calibration). Averaging does NOT remove it.
- **Gross error** = human blunder (misread 25.3 as 25.5, wrong button). Fixes by re-measuring carefully.

Q2. A rod is measured 25.60 cm; true value 25.30 cm. Find the three error numbers.

> *Three numbers, in this order, every time:*

> **Absolute error** $\Delta L = |\text{measured} - \text{true}| = |25.60 - 25.30| =$ **0.30 cm**

> **Relative error** $= \Delta L / \text{true} = 0.30 / 25.30 =$ **0.0119**

> **Percentage error** $= 0.0119 \times 100 =$ **1.19%**

> **Technique B — always divide by TRUE value, not measured. Absolute has units. Relative and % have none.**

Q3. A disc has radius r = (10.0 ± 0.2) cm. Find the % error in its area $A = \pi r^2$.

> **Technique C — the POWER RULE.** If a quantity is raised to power n, its relative error gets multiplied by n. The constant π carries no error.

> % error in A = n × (% error in r) = $2 \times (0.2/10.0) \times 100 = 2 \times 2\% =$ **4%**.

Q4. A = (50.0 ± 0.5), B = (20.0 ± 0.2). Find the error in Z = A/B.

> **Technique D — the OPERATION RULE.** Multiply/divide → ADD relative errors. Add/subtract → ADD absolute errors. Never mix the two.

> Relative error = 0.5/50.0 + 0.2/20.0 = 0.01 + 0.01 = 0.02 → **2%**. Z = 2.5, so $\Delta Z = 0.02 \times 2.5 = 0.05$ → **Z = 2.5 ± 0.05**.

Q5. A cube side (5.00 ± 0.02) cm. % error in volume, and the maximum absolute error?

> $V = s^3$. Power rule: % error = $3 \times (0.02/5.00) \times 100 = 3 \times 0.4\% =$ **1.2%**.

> Absolute: $\Delta V = 3s^2\Delta s = 3 \times 25 \times 0.02 =$ **1.5 cm³**.

Q6. Check whether $s = ut + \frac{1}{2}at^2$ is dimensionally correct.

> **Technique E — dimensional check.** Every term added must have the SAME dimension.

> $[s] = L$. $[ut] = LT^{-1} \cdot T = L$. $[at^2] = LT^{-2} \cdot T^2 = L$. $L = L + L$ ✓. Write the words "both terms have dimension L, equation dimensionally correct".

Q7. Write the dimensional formula of: velocity, acceleration, force, work, pressure.

> **Technique F — build from M, L, T.** Velocity = d/t = $\mathbf{LT^{-1}}$. Acceleration = velocity/t = $\mathbf{LT^{-2}}$. Force = mass×acceleration = $\mathbf{MLT^{-2}}$. Work = force×distance = $\mathbf{ML^2T^{-2}}$. Pressure = force/area = $\mathbf{ML^{-1}T^{-2}}$.

Q8. 0.000456 and 7,850,000 — write each in scientific notation and give its order of magnitude.

> $0.000456 = 4.56 \times 10^{-4}$ → order 10^{-4}. $7{,}850{,}000 = 7.85 \times 10^6$.

> **Technique G — order of magnitude.** Look at the leading digit. If it is 5 or more, round the power UP. $7.85 \geq 5 \rightarrow$ $\mathbf{10^7}$. If below 5, keep the power.

Q9 (his signature story question). Two teams send data in different unit systems and a spacecraft is lost. Explain what happened and what should be enforced.

> *Never panic on a story question. The story is just Technique A/B/C wearing clothes.*

> 1999 Mars Climate Orbiter: one team gave thrust in pound-force·seconds (Imperial), the navigation team expected newton·seconds (SI). The same equation mixed two unit systems, so the navigation model was wrong by a factor of about 4.45. The craft entered the Martian atmosphere too low and was destroyed.

> **Lessons to list (he awards marks per point):** enforce SI-only pipelines; tag every variable with its unit; use software that checks units automatically; run dimensional analysis as a sanity check; standardise interfaces between teams.

CHAIN 1 EXAM PREVIEW: the error-propagation 5-mark question (Q3–Q5 type), a dimensional check (2 marks), and either the Mars story or an MCQ picking random vs systematic. If you can do Q2–Q5 without notes, you own this block.

CHAIN 2 — VECTORS, DERIVATIVES, GRAPHS (40 min)

Q10. A scalar is multiplied by a negative number. What changes about the vector?

> Magnitude scales by $|k|$. **Direction reverses.** One-line answer, and it has been on his paper.

Q11. What is A·B, and what is it when the vectors are perpendicular?

> **Technique H — two products, two jobs.**

> **Dot product** $A \cdot B = AB\cos\theta$. Result is a SCALAR. Perpendicular → $\cos 90° = 0$, so $\mathbf{A \cdot B = 0}$.

> **Cross product** $A \times B = AB \sin\theta\, \hat{n}$. Result is a VECTOR, direction by the **right-hand rule**. Parallel → 0.

Q12. Find dy/dx of $y = 4x^2 - 3x + 7$.

> **Technique I — the derivative rule.** Multiply by the power, then drop the power by one. Constant terms die.

> $dy/dx = 4 \times 2x - 3 =$ $\mathbf{8x - 3}$.

Q13. Integrate $\int(3x^2 + 2x)\, dx$.

> **Technique J — the integration rule (reverse of I).** Raise the power by one, divide by the new power. Always add **+ C**.

> $\int(3x^2 + 2x)dx = 3 \cdot x^3/3 + 2 \cdot x^2/2 + C =$ **$x^3 + x^2 + C$**.

Q14. What does the slope of a position-time graph give? A straight line vs a rising curve?

> **Technique K — the graph ladder.** Position→velocity→acceleration: each graph's SLOPE gives the next quantity, each graph's AREA gives the one before.

> Straight rising line = **constant velocity**; average and instantaneous velocity are equal. Curved rising (steepening) = **velocity increasing**; the tangent slope grows with time.

Q15. Car A moves 20 m/s, Car B 12 m/s, same direction. Velocity of A relative to B? Opposite directions?

> **Technique L — relative velocity.** $v(A/B) = v_A - v_B$. Always subtract in that order.

> Same direction: 20 – 12 = **8 m/s**. Opposite: take B as –12, so 20 – (–12) = **32 m/s**.

CHAIN 2 EXAM PREVIEW: derivative/integration as a 2-mark short, graphs as a 2-mark short, relative velocity inside a 5-mark problem.

CHAIN 3 — STRAIGHT-LINE MOTION (30 min)

Q16. A body starts from rest, $a = 2$ m/s². Velocity after 5 s?

> **Technique M — pick the formula by what is missing.**

> $v = u + at$ (no distance) · $v^2 = u^2 + 2as$ (no time) · $s = ut + \frac{1}{2}at^2$ (no final velocity).

> $v = 0 + 2 \times 5 =$ **10 m/s**.

Q17. $a = 10t + 6$, with $v(0) = 4$ and $x(0) = 3$. Find $v(t)$, $x(t)$, and $x(1)$.

> **Technique N — integration chain: a → v → x.** Integrate once for v, twice for x. Pin the constants using the t = 0 values.

> $v(t) = \int a\,dt = 5t^2 + 6t + C$. $v(0) = 4 \rightarrow C = 4 \rightarrow$ **$v(t) = 5t^2 + 6t + 4$**.

> $x(t) = \int v\,dt = (5/3)t^3 + 3t^2 + 4t + C$. $x(0) = 3 \rightarrow C = 3 \rightarrow$ **$x(t) = (5/3)t^3 + 3t^2 + 4t + 3$**.

> $x(1) = 5/3 + 3 + 4 + 3 =$ **14.67 m**.

Q18. $x(t) = 3t^2 + 4t + 2$, $y(t) = 2t^3 - 3t^2 + 5t$ (a drone). Velocity, speed, acceleration, direction at t = 2 s.

> Same technique N, but on two axes, then combine.

> $v_x = 6t + 4 \rightarrow$ at t=2: **16**. $v_y = 6t^2 - 6t + 5 \rightarrow$ at t=2: **17**. Speed = $\sqrt{(16^2+17^2)} =$ **23.3 m/s**.

> $a_x = 6$. $a_y = 12t - 6 \rightarrow$ at t=2: **18**. $|a| = \sqrt{(6^2+18^2)} =$ **19.0 m/s²**. Direction $\theta = \tan^{-1}(17/16) =$ **46.7°** from +x.

Q19. A car goes 0 → 20 m/s in 5 s, then holds 20 m/s for 5 s. Displacement, average velocity, average acceleration in the first 5 s?

> **Technique O — v-t graph.** Area under the curve = displacement. Slope = acceleration.

> Displacement = area of triangle + area of rectangle = $\frac{1}{2} \times 5 \times 20 + 5 \times 20 =$ **100 m**. Average velocity = 100/10 = **10 m/s**. Average acceleration = (20–0)/5 = **4 m/s²**.

CHAIN 3 EXAM PREVIEW: a 3-mark v=u+at, a 5-mark differentiate-integrate, or the graph story.

CHAIN 4 — PROJECTILE (30 min)

Q20. A ball leaves the ground at 25 m/s, 37° to the horizontal. Components, time of flight, max height?

> **Technique P — the projectile four-step.** Split u, then T, then H. g is printed on the paper — READ IT (this one said g = 10, sin37 = 0.6, cos37 = 0.8).

> ux = 25 × 0.8 = **20 m/s**. uy = 25 × 0.6 = **15 m/s**.

> T = 2uy/g = 30/10 = **3.0 s**. H = $uy^2/2g$ = 225/20 = **11.25 m**.

> Range would be R = ux × T = 20 × 3 = **60 m**.

Q21. At what angle is the range maximum, and which other angle gives the same range?

> **45°**. Complementary angles (θ and 90−θ) give the same range.

Q22 (10-mark story type). An airplane points due north at 100 km/h. A wind blows east at 60 km/h. Answer four parts.

> **Technique Q — his four verbs.** He always asks in this order: (a) *Evaluate a claim*, (b) *Support your judgment*, (c) *Design a direction*, (d) *Assess a limitation*. Answer all four, labelled, even if part (d) is a guess.

> **(a) Claim:** "pointing north means it travels north over the ground." **Incorrect.** Ground velocity = velocity through air + wind velocity, so the plane also drifts east.

> **(b) Support:** v_ground = 100ĵ + 60î. Magnitude $\sqrt{(100^2+60^2)}$ = **116.6 km/h**. Direction θ = $\tan^{-1}(60/100)$ = **31° east of north**.

> **(c) Design:** to go due north, the east component must cancel the wind: point slightly west of north, with air-velocity east component −60 km/h. North component = $\sqrt{(100^2 - 60^2)}$ = **80 km/h** ground speed.

> **(d) Limitation:** if wind were 120 km/h and the plane can only do 100, cancellation is impossible — the craft is always blown east. Justify with components.

Q23 (same technique, new story). A person in the back of a truck throws a ball straight up at 5.0 m/s relative to the truck. The truck moves at 10 m/s and accelerates at 2 m/s². g = 10.

> **(a) Claim:** "the ball returns to my hand because it already has the truck's horizontal velocity." **Incorrect.** The ball keeps the truck's velocity at the instant of release, but the truck then accelerates forward, so it pulls ahead.

> **(b) Support:** T = 2 × 5.0 / 10 = **1.0 s**. Ball travels 10 × 1 = 10 m. Hand travels 10×1 + ½×2×1^2 = 11 m. Ball is **1 m behind the hand**.

> **(c) Design:** to land back in the hand, the ball needs an extra 1 m in 1 s → **+1 m/s forward relative to the truck**. If the truck's speed doubled with the same acceleration, the extra distance is still ½aT^2 = 1 m, so the required toss is **unchanged**.

> **(d) Check:** the time 1.0 s comes from 5.0 and 10.0, both 2 significant figures → answer **2 significant figures**.

CHAIN 4 EXAM PREVIEW: the projectile 5-mark (Q20) is near-certain. Learn Technique P by writing it out twice.

CHAIN 5 — FORCES AND FRICTION (45 min)

Q24. 10 kg pulled by 35 N on a floor, μk = 0.20. Acceleration? (g = 9.8)

> **Technique R — the force ladder.** (1) draw the free-body diagram (2) find the normal force N (3) find friction (4) net force (5) a = F/m.

> N = mg = 98 N. f_k = μk N = 0.20 × 98 = 19.6 N. F_{net} = 35 − 19.6 = 15.4 N. a = 15.4/10 = **1.54 m/s²**.

Q25. 6 kg, μs = 0.50, μk = 0.30, applied force 20 N. Does it move?

> f_s,max = μs mg = 0.50 × 6 × 9.8 = 29.4 N. Since 20 < 29.4, it **does not move**. Static friction matches the applied force exactly: **20 N**.

> **Technique S — the static trick.** Static friction is self-adjusting, up to a maximum. It is NOT always μsN.

Q26. 5 kg, 40 N applied at 30° ABOVE the horizontal, μk = 0.20.

> Pulling up reduces the normal force.

> Fx = 40 cos30 = 34.64 N. Fy = 40 sin30 = 20 N. N = mg − Fy = 49 − 20 = **29 N**. f_k = 0.20 × 29 = 5.8 N. a = (34.64 − 5.8)/5 = **5.77 m/s²**.

Q27. 10 kg, 50 N applied at 30° BELOW the horizontal (pushing), μk = 0.25.

> Pushing down increases the normal force.

> N = mg + 50 sin30 = 98 + 25 = **123 N**. f_k = 0.25 × 123 = 30.75 N. a = (50 cos30 − 30.75)/10 = (43.30 − 30.75)/10 = **1.26 m/s²**.

> **Technique T — compare Q26 and Q27.** Pull up → N smaller, a larger. Push down → N larger, a smaller. He loves this pair.

Q28. 5 kg rests on a 37° incline, μs = 0.50. Does it slide? (g = 10)

> **Technique U — the incline test.** Sliding component = mg sinθ. Resisting maximum = μs mg cosθ. Compare them. The quick version: slides iff tanθ > μs.

> tan37 = 0.75 > 0.50 → **it slides**. (mg sin37 = 30 N vs μs mg cos37 = 20 N.)

Q29. 10 kg slides down a 30° incline, μk = 0.20.

> a = g(sinθ − μk cosθ) = 9.8(0.5 − 0.20×0.866) = 9.8(0.5 − 0.173) = **3.20 m/s²** down the plane.

Q30 (the pulley classic). 5 kg on a rough table (μk = 0.20), tied to a hanging 3 kg over a frictionless pulley.

> **Technique V — whole-system acceleration.** Divide the driving force by the TOTAL mass.

> Driving = $m_2 g$ = 3 × 9.8 = 29.4 N. Resisting = μk m_1 g = 0.20 × 5 × 9.8 = 9.8 N. Total mass = 8 kg.

> a = (29.4 − 9.8)/8 = **2.45 m/s²**. Tension T = $m_2(g - a)$ = 3 × 7.35 = **22.05 N**. Check with the table block: T − f_k = 22.05 − 9.8 = 12.25 = 5 × 2.45 ✓

CHAIN 5 EXAM PREVIEW: friction on an incline or a pulley is his favourite 5-mark. Write the free-body diagram first — he draws them in his own notes and rewards the diagram.

CHAIN 6 — CIRCULAR MOTION (25 min)

Q31. A car turns in a circle of radius 50 m at 10 m/s. Angular velocity and centripetal acceleration?

> **Technique W — the circular set.** ω = v/r. a_c = v²/r = rω². Period T = 2πr/v.

> ω = 10/50 = **0.2 rad/s**. a_c = 10²/50 = **2 m/s²**, directed toward the centre.

Q32. A wheel of radius 0.5 m spins at 120 rpm. Linear speed?

> Convert first: ω = 120 × 2π/60 = 4π rad/s. v = rω = 0.5 × 4π = **6.28 m/s**.

Q33. Torque from a 100 N force at r = 0.5 m, when the force is at 90°, 60°, 0° to the radius?

> τ = rF sinθ. 90°: 0.5 × 100 × 1 = **50 N·m**. 60°: **43.3 N·m**. 0°: **0**.

> **Technique X — the zero-torque rule.** A force along the radius does no torque, exactly like the centripetal force does no work.

Q34. Why does the centripetal force do no work?

> It is always perpendicular to the displacement, and W = Fd cos90° = 0. Therefore it changes direction only, never speed.

CHAIN 6 EXAM PREVIEW: usually a 3-mark ω/a_c calculation plus one "why is it zero" short.

CHAIN 7 — WORK, ENERGY, CENTRE OF MASS (35 min)

Q35. A 4 kg block is pushed by 10 N over 3 m, with friction 2 N. Work by push, by friction, net work, final speed from rest?

> **Technique Y — work–energy theorem.** $W_{net} = \Delta KE = \frac{1}{2}mv^2$.

> $W_{push} = 10 \times 3 = 30$ J. $W_{friction} = -2 \times 3 = -6$ J. $W_{net} =$ **24 J**.

> $24 = \frac{1}{2} \times 4 \times v^2 \rightarrow v^2 = 12 \rightarrow v =$ **3.46 m/s**.

Q36. Which forces are conservative, and how does that change the energy?

> **Technique Z — conservative vs non-conservative.** Gravity and spring force are conservative: work depends only on start and end points, and a closed loop gives zero work, so KE + PE is constant. Friction is non-conservative: work depends on the path and always removes mechanical energy, turning it into heat, so total mechanical energy is NOT conserved.

Q37. 2 kg at x = 0 and 8 kg at x = 5. Centre of mass?

> **Technique AA — weighted average.** $x_{CM} = \Sigma mx / \Sigma m = (2 \times 0 + 8 \times 5)/10 =$ **4 m**. The centre of mass always sits closer to the heavier mass.

Q38. 2 kg at (0,0), 3 kg at (6,0), 5 kg at (0,4). Centre of mass?

> Do each axis separately. $x_{CM} = (0 + 18 + 0)/10 =$ **1.8**. $y_{CM} = (0 + 0 + 20)/10 =$ **2.0**. Answer: **(1.8, 2.0)**.

Q39 (his contest question, still unanswered, 0/40 XP). An ice cube of mass m melts in a tray of mass M. Does the centre of mass move, in which direction, and what sets the size of the move?

> **Technique AB — the reasoning script (no numbers wanted).**

> 1. Does it move? Yes.

> 2. Why? The centre of mass depends on how mass is distributed vertically. While solid, the ice's mass sits at height L/2. After melting, the water lies in a thin layer at the tray base.

> 3. Direction? The mass is redistributed downward, so the combined centre of mass **drops toward the base**.

> 4. What sets the size? The drop is about $(mL/2)/(M+m)$. It is larger when the ice is heavy compared with the tray ($m \gg M$) and smaller when the tray is heavy ($M \gg m$).

> **Template for any variant:** claim → distribution change → direction → mass ratio controls the size.

CHAIN 7 EXAM PREVIEW: a 5-mark work–energy or a CoM calculation, and possibly the ice-tray reasoning as a long answer. Learn the template, not the ice cube.

CHAIN 8 — MOMENTUM AND COLLISIONS (45 min)

Q40. A 4 kg ball at 6 m/s hits a stationary 2 kg ball and they stick. Common velocity?

> **Technique AC — the momentum rule.** If the net external force is zero, total momentum before = total momentum after. Write it as two sides of an equation, every time.

> $4\times6 + 0 = (4+2)v \rightarrow v =$ **4 m/s**.

Q41. A 3 kg ball at 6 m/s hits a stationary 2 kg ball with restitution e = 0.4. Both final velocities?

> **Technique AD — the two-equation machine.** (1) momentum: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$. (2) restitution: $v_2 - v_1 = e(u_1 - u_2)$, meaning separation speed = e × approach speed.

> $3v_1 + 2v_2 = 18$ and $v_2 - v_1 = 2.4$. Solve: $\mathbf{v_1 = 2.64}$ **m/s,** $\mathbf{v_2 = 5.04}$ **m/s**.

Q42. What does e = 1 mean? e = 0? Two identical balls, one elastic, one at rest?

> e = 1 is perfectly elastic (kinetic energy also conserved). e = 0 is perfectly inelastic (they stick). For two identical masses with e = 1 and the target at rest, the velocities are **exchanged**: the first stops, the second takes all the speed.

Q43. A 0.5 kg ball hits a wall at 10 m/s and rebounds at 6 m/s. Initial momentum, final, change, impulse?

> $p_i = 0.5 \times 10 =$ **5**. $p_f = 0.5 \times (-6) =$ **–3**. $\Delta p = -3 - 5 =$ **–8 kg·m/s**. Impulse = Δp = **–8 N·s**, opposite to the original motion.

> **Technique AE — pick a positive direction first and write it on the diagram.** The minus signs then take care of themselves.

Q44. A 10 kg body at rest explodes into 4 kg at 12 m/s and a 6 kg piece. Velocity of the second piece?

> Internal forces only, momentum still conserved. $0 = 4\times12 + 6v \rightarrow v =$ **–8 m/s**, meaning 8 m/s in the opposite direction.

Q45. 2 kg east at 6 m/s and 3 kg north at 4 m/s. Components of total momentum, magnitude, direction?

> **Technique AF — 2D momentum.** Conserve or add each axis separately, then combine with Pythagoras.

> $p_x = 12$, $p_y = 12$. $|p| = \sqrt{(12^2+12^2)} =$ **17.0 kg·m/s**. $\theta = \tan^{-1}(12/12) =$ **45° north of east**.

Q46. 2 kg east at 5 m/s and 1 kg north at 4 m/s stick together. Final velocity and direction?

> $p_x = 10$, $p_y = 4$, total mass 3. $v_x = 10/3$, $v_y = 4/3$. $v = \sqrt{(100+16)}/3 =$ **3.57 m/s**. $\theta = \tan^{-1}(4/10) =$ **21.8° north of east**.

Q47. What does $F_{net} = 0$ on a whole system tell you about three separate things?

> Total momentum is **constant**. The centre-of-mass velocity is **constant**. The centre-of-mass acceleration is **zero**, and $a_{cm} = F_{net}/M$.

Q48. 2 kg at 4 m/s and 3 kg at 6 m/s, same direction. Total momentum and centre-of-mass velocity?

> $p = 8 + 18 =$ **26 kg·m/s**. $v_{cm} = 26/5 =$ **5.2 m/s**.

CHAIN 8 EXAM PREVIEW: collisions are his most reliable 5-mark. Technique AD (two equations) is worth two marks on its own if you only write them down.

MASTERY TEST (20 min, tomorrow 7 AM, no notes)

Answer out loud in one sentence each. The bracket is the technique.

1. Difference between random and systematic error? (A)
2. % error in $A = \pi r^3$ with $r \pm 2\%$? (C)
3. Derivative of $y = 5x^3 - 2x$? (I)
4. Velocity after 5 s from rest at $a = 3\ m/s^2$? (M)
5. Range maximum at what angle? (P)
6. Does static friction always equal $\mu s N$? (S)
7. Incline slides when $\tan\theta$ is greater or less than μs? (U)
8. Pulley acceleration: driving force over what mass? (V)
9. Why does centripetal force do no work? (X)
10. Conservative force on a closed loop? (Z)
11. CoM of 2 kg at 0 and 8 kg at 10? (AA)
12. Momentum conserved when what is zero? (AC)
13. e = 1 and e = 0? (AD)
14. Explosion: total momentum before? (AE, it is zero)
15. 2D momentum: combine axes how? (AF)
16. Ice melts: which way does the CoM move? (AB)

14–16 correct: you are comfortably above the floor. 11–13: fine. Under 10: re-open only those technique boxes.

EXAM-DAY PROTOCOL

1. **First pass, 35 min:** all five 3-mark questions. These are definition and one-step calculation questions from the technique boxes. Never leave one blank — write the formula and one sentence.
2. **Second pass, 60 min:** the three 5-mark questions. Use the (A)/(B) choice to pick the pair whose technique you can write from memory. Label the sub-parts (a)(b)(c).
3. **Last 15 min:** the 10-mark story question. Use Technique Q: (a) claim, (b) calculate, (c) design, (d) limitation. Four labelled paragraphs beats one long one.
4. **If a question looks new:** it is one of his worksheets with different numbers. Name the chain, pull the technique, re-run the numbers.
5. **g:** read it from the paper. Worksheets used 9.8, one contest used 10. sin37 = 0.6, cos37 = 0.8 when needed.
6. **Three-minute rule:** stuck more than three minutes? Write the formula and what you do know, then move on and return at the end.
7. **Marks live in method.** Show every step on a new line with units. His papers reward written justification ("because", "since the net force is zero") over bare answers.

All the best tomorrow.